

Snap2Cal: A Food Calorie Tracker Application

Anonymous submission

Abstract

This report presents a deep learning-based food classification application designed to identify food items from images and retrieve their nutritional information for dietary tracking applications. It utilizes the MobileNetV2 model, the system was trained on the Food-101 dataset, encompassing 101 food categories, and integrated with USDA FoodData Central nutritional data via a Flask API. Initial experiments reveal a macro-average F1-score of 0.58, indicating moderate performance with notable variability across categories. The system aims to support health-conscious users by providing accurate food identification and nutritional insights, with potential applications in mobile health platforms. The authors made the model novel and improved it to fix the flaws that the model had. The goal was to create a model that uses a convolutional neural network to train with the Food101 dataset, properly label it, and give the nutritional facts of that food item.

Keywords: Food-101, Convolutional Neural Network, iOS, Computer vision, USDA, Mobile Application

Introduction

The increasing prevalence of dietary-related health issues, such as obesity and diabetes, underscores the need for automated tools to assist with food intake monitoring. Traditional methods, such as manual food logging, are time-consuming and prone to errors, limiting their effectiveness for health management. This project addresses these challenges by developing a food classification system that uses deep learning to identify food items from images and retrieve their nutritional profiles. Leveraging the Food-101 dataset (Bossard, Guillaumin, and Van Gool 2014), which includes 101 food categories and 101,000 images, the system employs the MobileNetV2 model for efficient image classification (Sandler et al. 2018). Nutritional data is sourced from the USDA FoodData Central database (USDA FoodData Central 2025), integrated via a Flask API for real-time access. The primary objective is to create a scalable, accurate tool for dietary tracking, with potential applications in mobile health platforms. This work contributes to the field by combining advanced image recognition with nutritional data integration, addressing both classification accuracy and practical utility for health-conscious users.

Related Work

Early Forays into Food Image Classification

Food image classification has developed at considerable pace in the past decade, made possible through the advent of deep learning techniques. Kagaya et al. (Kagaya, Aizawa, and Ogawa 2014) were the first to try food image classification, employing convolutional neural networks (CNNs). Their approach demonstrated the ability of CNNs in food recognition with promising results on relatively small datasets. But the limitations of model complexity and dataset size in these early works did not allow the building of useful food classification systems for real-world use.

Integration of Nutritional Data with Food Classification

Recent advances have extended food image classification into nutrition-orientated information, thus extending the practical applicability of such systems to monitoring diets. For example, the Calorie Mama application (Calorie Mama 2025) blends food image classification with an offboard database of nutrition data to inform users with real-time dietary analysis. This implies the prospect of unifying image-based classification with offboard data sources, although the challenge of achieving optimization of both accuracy and real-time operation on mobile platforms remains.

The Food-101 Dataset and State-of-the-Art Architectures

The Food-101 dataset (Bossard, Guillaumin, and Van Gool 2014) is one of the high-capacity benchmarks for food image classification and has been used for comparing various deep learning architectures in extensive studies. Of all state-of-the-art models that have been tested over this dataset, EfficientNet (Tan and Le 2019) has reported top-1 accuracies of approximately 82%, reflecting a trade-off between computational efficiency and classification accuracy. Yet, while such models perform well for classifying food images, they usually don't take into account the incorporation of nutritional information, which is paramount to applications like dietary tracking and personal nutrition.

Efficient Classification with MobileNetV2

Here, we extend prior work by employing MobileNetV2 (Sandler et al. 2018), a lightweight convolutional network model, to implement rapid food image classification. MobileNetV2's reduced computational cost positions it particularly ideally suited for resource-constrained devices such as smartphones, a consideration highly relevant to real-time food identification applications. Coupled with the USDA's National Nutrient Database (USDA FoodData Central 2025), this paper suggests combining accurate image classification with nutritional analysis in order to offer an end-to-end solution to monitoring diets.

Large-Scale Datasets for Comprehensive Food Classification

One major contribution to the field of food image classification research is the AI4Food-NutritionDB GitHub repository, which combines a broad range of food image datasets into one large-scale resource. The repository consists of over 500,000 images from seven food image datasets, providing a comprehensive collection for training strong food classification models. The datasets employed are:

- UECFood-256 (of Electro-Communications n.d.),
- Food-101 (Bossard, Guillaumin, and Van Gool 2014),
- Food-11 (Authors n.d.a),
- FruitVeg-81 (Authors n.d.b),
- MAFood-121 (Authors n.d.d),
- ISIA Food-500 (Authors n.d.c),
- VIPER-FoodNet (Authors n.d.e).

While this research works with a sample of these datasets, the AI4Food-NutritionDB repository speaks to the incredible efforts being put into improving the accuracy and range of food classification systems. The ability to work with different datasets across the world enables more generalized models to be created that can recognize more diverse foods, thereby making them more applicable for global diet tracking systems.

Methodologies

The proposed system consists of two main components: a deep learning model for food image classification and a nutritional data retrieval pipeline. The MobileNetV2 model, known for its efficiency in mobile applications

Convolutional Neural Networks (CNNs) are a specialized type of deep learning neural network designed to process and analyze visual data. They utilize a series of layers to extract and interpret features from images. The core component, the convolution layer, applies filtering operations to identify patterns.

Compared to traditional neural networks, CNNs excel in handling image, audio, video, and speech signals due to their enhanced performance. CNNs typically consist of three key layer types: convolutional, pooling, and fully connected layers. The convolutional layer serves as the initial layer, followed by additional convolutional or pooling layers, with the fully connected layer concluding the network.

With each layer, the CNN grows more complex, detecting increasingly intricate aspects of the image. Early layers focus on fundamental features like edges and colors, while deeper layers identify larger components or shapes, ultimately recognizing the intended object.

The convolutional layer is the cornerstone of CNNs, as it handles the bulk of the computational workload. It processes a set of images to produce a feature map. For instance, a grayscale image is represented as a 2D pixel matrix, with input dimensions encompassing height and width. A feature detector—often called a kernel or filter—scans the image to identify specific features. While filter sizes may vary, a 3x3 matrix is frequently used. The outcome of these convolution operations is a feature map, also referred to as an activation map or convolved feature.

(Sandler et al. 2018), was selected as the base architecture. It was pre-trained on ImageNet and fine-tuned on the Food-101 dataset, which comprises 101,000 images across 101 food categories (Bossard, Guillaumin, and Van Gool 2014). The dataset was split into 75,750 training images (750 per class), 12,625 validation images (125 per class), and 12,625 test images (125 per class). Data augmentation techniques, including rotation (up to 40 degrees), width/height shifts (0.3), shear (0.3), zoom (0.3), horizontal/vertical flips, brightness adjustments (0.7-1.3), and channel shifts (20.0), were applied to enhance model generalization.

The model architecture includes MobileNetV2's base layers, followed by a global average pooling layer, a dense layer with 512 units and ReLU activation, a dropout layer (0.5), and a final dense layer with 101 units and softmax activation for classification. Label smoothing was applied to the categorical cross-entropy loss to mitigate overconfidence. The model was initially trained for 20 epochs with the base layers frozen, using an Adam optimizer (learning rate 0.001), followed by fine-tuning for 10 epochs with the last 40 layers unfrozen (learning rate 1e-5). A learning rate scheduler reduced the rate by a factor of 0.5 if validation loss plateaued for 2 epochs, with a minimum of 1e-6. Early stopping (patience=5) ensured training stopped if validation loss did not improve.

MobileNetV2 is a lightweight deep learning model, built on the principles of CNNs, specifically designed for resource-constrained devices like mobile phones and embedded systems. It excels in tasks such as image classification, object detection, and semantic segmentation by efficiently processing visual data with minimal computational cost. Unlike traditional CNNs, MobileNetV2 optimizes performance through a streamlined architecture that balances accuracy and efficiency, making it ideal for real-time applications on low-power devices.

MobileNetV2 processes images similarly to a CNN but with optimizations for efficiency. The input image, represented as a 3D matrix (height, width, and color channels), is fed into the network. The initial convolutional layer generates a feature map by applying filters to detect low-level features like colors and edges. As the data passes through the inverted residual blocks, the network becomes more complex, identifying increasingly abstract features.

TODO - We need to add here what we did to make our model unique and our own

Nutritional data was sourced from the USDA FoodData Central database (USDA FoodData Central 2025) and pre-processed into 'food101_usda_nutrition.csv', mapping each Food-101 category to nutritional metrics (calories, protein, fat, carbs). A Flask API was developed to handle image uploads, run the TFLite model for classification, and retrieve nutritional data based on the predicted class. The API is deployed on a local server, accessible at 'http://localhost:5000/classify'.

Experiments and Results

Experiments were conducted using the Food-101 test set (12,625 images, 125 per class). The model's performance was evaluated using precision, recall, and F1-score, with results aggregated into macro and weighted averages.

The CNN was connected to the iOS app we made with a TFLite file. A TFLite file is a machine learning model used by TensorFlow Lite. The model contained in a TFLite file is used for image classification. This allowed the model to detect what food item is in the image. Every time the model was trained, a new TFLite was created and then was used to replace the old TFLite connected to the app. The app works very intuitively with the Python code created. It runs on a flask server to work with the TFLite and the USDA CSV that has the nutritional facts.

Due to space constraints, the classification report is split into two tables (Table 1 and Table 2), revealing a macro-average F1-score of 0.58, indicating moderate overall performance. Accuracy across all classes is 0.59, reflecting challenges in distinguishing visually similar foods.

Categories like edamame (F1-score: 0.94) and oysters (0.82) performed well, likely due to distinct visual features, while visually similar items, such as apple pie (0.26) and pancakes (0.60), showed confusion, with apple pie images often misclassified as pancakes. This suggests the model struggles with fine-grained differentiation, possibly due to limited training epochs (10 initial, 5 fine-tuning) and insufficient feature learning for overlapping categories.

Conclusions

This project demonstrates a viable deep learning approach for food classification and nutritional data retrieval, achieving a macro-average F1-score of 0.58. While categories with distinct visual features performed well, the model requires improvement for visually similar foods. Future work could involve increasing training epochs, fine-tuning more layers, or exploring advanced architectures like EfficientNet to enhance classification accuracy.

It is novel to create an app from scratch and then combine it with a conventional neural network. While it was prebuilt, the authors took it, tweaked it, and led to a working food app. This shows just how machines are capable of doing only what the mind sets itself to. This idea can be put forward to help support other apps like MyFitnessApp or become a new app that pushes the boundaries of what food classifiers do.

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Dish	Precision	Recall	F1-Score	Support
apple_pie	0.41	0.19	0.26	250
baby_back_ribs	0.56	0.63	0.59	250
baklava	0.53	0.67	0.59	250
beef_carpaccio	0.63	0.66	0.65	250
beef_tartare	0.57	0.53	0.55	250
beet_salad	0.44	0.39	0.41	250
beignets	0.69	0.81	0.75	250
bibimbap	0.76	0.80	0.78	250
bread_pudding	0.32	0.24	0.28	250
breakfast_burrito	0.47	0.52	0.50	250
bruschetta	0.55	0.42	0.47	250
caesar_salad	0.56	0.68	0.61	250
cannoli	0.64	0.62	0.63	250
caprese_salad	0.52	0.57	0.54	250
carrot_cake	0.59	0.50	0.55	250
ceviche	0.37	0.24	0.29	250
cheese_plate	0.58	0.62	0.60	250
cheesecake	0.51	0.32	0.40	250
chicken_curry	0.48	0.42	0.45	250
chicken_quesadilla	0.62	0.62	0.62	250
chicken_wings	0.54	0.73	0.62	250
chocolate_cake	0.51	0.59	0.55	250
chocolate_mousse	0.36	0.37	0.37	250
churros	0.72	0.73	0.73	250
clam_chowder	0.72	0.78	0.75	250
club_sandwich	0.65	0.66	0.66	250
crab_cakes	0.46	0.33	0.39	250
creme_brulee	0.71	0.75	0.73	250
croque_madame	0.60	0.64	0.62	250
cup_cakes	0.69	0.80	0.74	250
deviled_eggs	0.71	0.71	0.71	250
donuts	0.70	0.68	0.69	250
dumplings	0.78	0.84	0.81	250
edamame	0.91	0.96	0.94	250
eggs_benedict	0.56	0.78	0.65	250
escargots	0.78	0.71	0.74	250
falafel	0.65	0.47	0.54	250
filet_mignon	0.39	0.44	0.41	250
fish_and_chips	0.49	0.71	0.58	250
foie_gras	0.28	0.33	0.30	250
french_fries	0.66	0.81	0.73	250
french_onion_soup	0.66	0.75	0.70	250
french_toast	0.47	0.46	0.47	250
fried_calamari	0.62	0.58	0.60	250
fried_rice	0.72	0.62	0.67	250
frozen_yogurt	0.72	0.78	0.75	250
garlic_bread	0.57	0.46	0.51	250
gnocchi	0.42	0.40	0.41	250
greek_salad	0.46	0.65	0.54	250
grilled_cheese_sandwich	0.44	0.40	0.42	250
grilled_salmon	0.41	0.38	0.40	250

Table 1: Classification report for the initial MobileNetV2 model on the Food-101 test set (Part 1: apple_pie to grilled_salmon).

Dish	Precision	Recall	F1-Score	Support
guacamole	0.69	0.82	0.75	250
gyoza	0.56	0.66	0.61	250
hamburger	0.52	0.68	0.59	250
hot_and_sour_soup	0.70	0.86	0.78	250
hot_dog	0.72	0.76	0.74	250
huevos_rancheros	0.39	0.19	0.25	250
hummus	0.62	0.26	0.37	250
ice_cream	0.59	0.45	0.51	250
lasagna	0.54	0.49	0.51	250
lobster_bisque	0.69	0.72	0.71	250
lobster_roll_sandwich	0.66	0.63	0.64	250
macaroni_and_cheese	0.62	0.60	0.61	250
macarons	0.85	0.89	0.87	250
miso_soup	0.73	0.84	0.78	250
mussels	0.71	0.80	0.75	250
nachos	0.53	0.44	0.48	250
omelette	0.57	0.28	0.37	250
onion_rings	0.75	0.84	0.79	250
oysters	0.78	0.88	0.82	250
pad_thai	0.61	0.82	0.70	250
paella	0.62	0.56	0.59	250
pancakes	0.56	0.64	0.60	250
panna_cotta	0.51	0.49	0.50	250
peking_duck	0.47	0.64	0.54	250
pho	0.65	0.88	0.75	250
pizza	0.78	0.76	0.77	250
pork_chop	0.33	0.23	0.27	250
poutine	0.65	0.70	0.67	250
prime_rib	0.50	0.72	0.59	250
pulled_pork_sandwich	0.48	0.40	0.43	250
ramen	0.56	0.53	0.55	250
ravioli	0.34	0.18	0.23	250
red_velvet_cake	0.80	0.74	0.77	250
risotto	0.51	0.45	0.48	250
samosa	0.45	0.62	0.52	250
sashimi	0.74	0.86	0.79	250
scallops	0.41	0.42	0.42	250
seaweed_salad	0.76	0.78	0.77	250
shrimp_and_grits	0.41	0.47	0.44	250
spaghetti_bolognese	0.77	0.67	0.71	250
spaghetti_carbonara	0.80	0.80	0.80	250
spring_rolls	0.55	0.69	0.61	250
steak	0.32	0.25	0.28	250
strawberry_shortcake	0.55	0.62	0.58	250
sushi	0.64	0.56	0.60	250
tacos	0.43	0.35	0.39	250
takoyaki	0.71	0.55	0.62	250
tiramisu	0.51	0.42	0.46	250
tuna_tartare	0.37	0.22	0.28	250
waffles	0.75	0.64	0.69	250
accuracy	-	-	0.59	25250
macro avg	0.58	0.59	0.58	25250
weighted avg	0.58	0.59	0.58	25250

Table 2: Classification report for the initial MobileNetV2 model on the Food-101 test set (Part 2: guacamole to waffles, with summary).